

Freight Rate Prediction — Assessment Report

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1. Overview

The goal is to predict the posted rate for 12,000 freight loads dated November–December 2025, using 48,000 labeled loads from January–October 2025, plus a fixed 31-day December forecast for the Lexington → Fort Wayne lane. Because the prediction window lies entirely after the training window, I treated this as a **future forecasting problem**, which drove the validation design below.

2. Data Quality & Cleaning

- **Negative weights:** the weight column contained sign errors (e.g. −47,500 lb). Fixed with absolute value; ~300 missing weights imputed with the median weight per equipment type.
- **Missing market index:** 374 nulls in train, 249 in validation — imputed with the same-date median (month-level fallback).
- **Rate outliers:** posted rates ranged \$57–\$25,533 (0.33–14.1 \$/mile). Validation rows can never be dropped, so instead of filtering, the model was trained on a **log-transformed target** (log1p), which suppresses outlier influence while keeping every row.

3. Train/Validation Split

A random split would be inappropriate: it leaks future information into training and produces optimistic metrics that do not reflect the real task. The real task requires predicting **two months beyond** the training data (Nov–Dec from Jan–Oct), so the holdout was designed to mimic exactly that gap:

- **Train:** 2025-01-01 → 2025-08-31
- **Temporal holdout:** 2025-09-01 → 2025-10-31 (a two-month future window, matching the real Nov–Dec gap)

The final model was then retrained on the full January–October data (using the iteration count selected via early stopping on the holdout) before generating the November–December predictions. Imputers and encoders were fit on the training split only — no leakage.

Holdout results (Sep–Oct 2025):

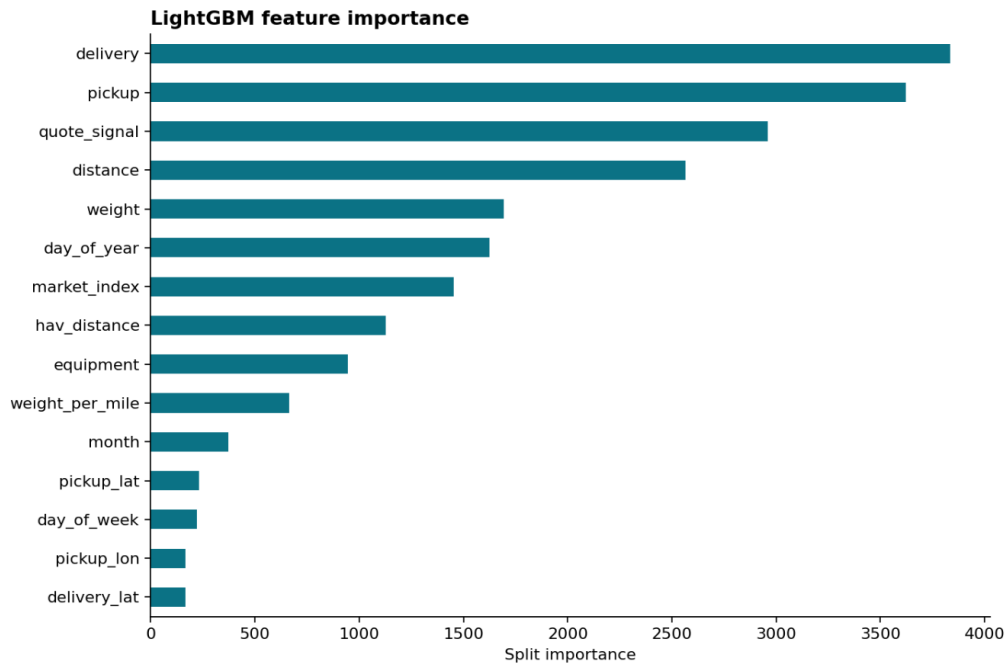
Model	MAE (\$)	MAPE (%)	RMSE (\$)
Baseline — equipment median \$/mile × distance	229.10	10.49	670.66
LightGBM (log target)	110.54	4.72	635.28

LightGBM cuts MAPE from 10.5% to **4.7%** — less than half the baseline error. RMSE improves less than MAE because a small number of extreme-rate loads dominate the squared error, which is expected in freight pricing data.

4. Model

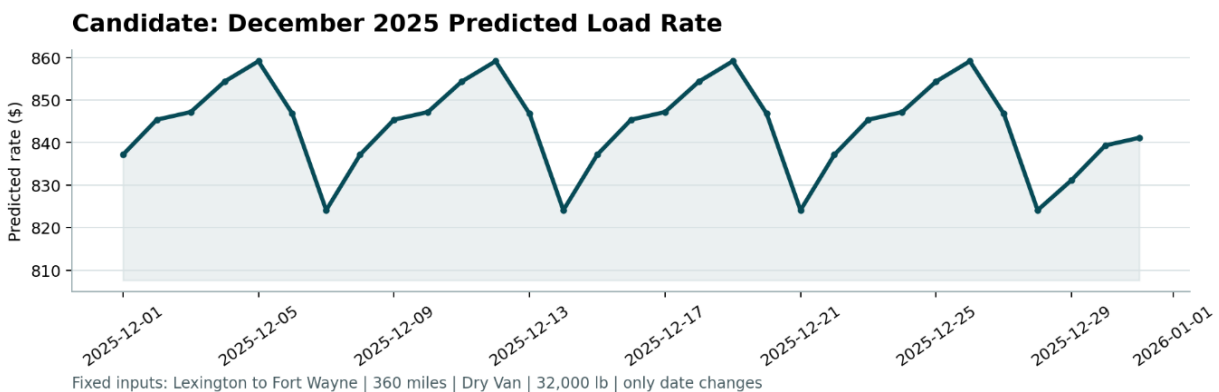
LightGBM gradient-boosted trees on the log-transformed rate. The data is tabular with categorical features (pickup, delivery, equipment) and non-linear interactions; gradient boosting handles both natively

and is the standard for pricing problems of this type. Features: distance, haversine distance from coordinates, equipment, cleaned weight, weight-per-mile, market index, quote signal, and calendar features (month, day-of-week, day-of-year). Early stopping on the temporal holdout controls overfitting; seeds are fixed for reproducibility. Feature importance confirms lane (pickup/delivery), quote signal, and distance as the strongest drivers.



5. Fixed December Predictions

The December input file contains only the seven base columns — no coordinates, market index, or quote signal. These were reconstructed: coordinates looked up from Lexington/Fort Wayne rows in training data; market index extrapolated from the observed winter seasonal pattern; quote signal imputed from comparable Dry Van loads (~360 mi). The chart below was generated by the provided score.py after validating all 12,000 predictions and the 31 December rows.



The model produces a clear weekly cycle (\$824–\$859): rates build through the week, peak midweek–Friday, and dip on weekends — consistent with the day-of-week pattern learned from ten months of training data.